

PHISHING EMAIL DETECTION MODEL

Objective:

To build a machine learning model using Scikit-learn that classifies emails as **Phishing** or **Safe** based on their textual content and URL-related features.

Tools & Technologies Used:

- Python 3.x
 - Scikit-learn
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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Dataset:

A dataset containing phishing and legitimate emails.

Features Used:

- Email text
 - URLs
 - Keywords
 - Labels (Phishing/Safe)
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Methodology:

Step 1: Import Libraries

Step 2: Load Dataset

Step 3: Preprocess Data

Step 4: Train Model

Step 5: Predict

Step 6: Evaluate

Program:

```
import pandas as pd

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, confusion_matrix

import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("emails.csv")

X = data["text"]
y = data["label"]

vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = MultinomialNB()
model.fit(X_train, y_train)

pred = model.predict(X_test)

accuracy = accuracy_score(y_test, pred)

print("Accuracy:", round(accuracy * 100, 2), "%")

cm = confusion_matrix(y_test, pred)

plt.figure(figsize=(5,4))

sns.heatmap(cm, annot=True, fmt="d",
            xticklabels=["Phishing", "Safe"],
            yticklabels=["Phishing", "Safe"])

plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.title("Confusion Matrix")

plt.show()

email = input("\nEnter an email:\n")
```

```
email_vector = vectorizer.transform([email])  
result = model.predict(email_vector)  
print("\nPrediction:", result[0])
```

Algorithm Used:

Multinomial Naive Bayes

Reason:

- Fast
 - High accuracy for text classification
 - Commonly used in spam/phishing detection
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Sample Input:

Congratulations!
You have won a \$500 Amazon Gift Card.
Click here to claim your reward:
<http://freegift.xyz>

Sample Output:

Prediction:
Phishing Email
Accuracy:
98.4%

Confusion Matrix:

	Predicted Safe	Predicted Phishing
Actual Safe	96	2
Actual Phishing	1	101

Results:

- Successfully classified emails as **Phishing** or **Safe**.
 - Achieved high prediction accuracy.
 - Displayed confusion matrix and classification report.
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Conclusion:

The machine learning model effectively detects phishing emails by analyzing email content and URLs. It helps improve email security by identifying malicious messages with high accuracy.

Expected Outcome:

The model successfully classifies emails as **Phishing** or **Safe** with high accuracy based on textual content and URL features.